

LIST OF PROBLEMS ABOUT KNOWLEDGE

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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1. I love Juliet, and I know that Juliet loves me, but I do not know if Juliet knows that I love her. Use a knowledge space this situation, and specify the state of the world in that knowledge space that corresponds to the situation described.
2. Romeo composes a letter to Juliet, and gives it to Tybalt to deliver to Juliet. While on the way, Tybalt peeks at the letter's contents. Tybalt gives Juliet the letter, and Juliet reads it immediately, in Tybalt's presence. Neither Romeo nor Juliet knows that Tybalt has read the letter. Answer the following questions about this story:
 - (a) Provide a knowledge space in which all the elements of the story above regarding the knowledge possessed by Romeo, Tybalt, and Juliet regarding the content of the letter hold true. Specify the state of the world in the model that corresponds to the situation described above.
 - (b) In the state of the world you specified above, does Romeo know that Juliet has read the letter? Justify.
 - (c) In the state of the world you specified above, does Tybalt know that Romeo knows that Juliet has read the letter? Justify.
 - (d) Construct a knowledge space in which, in addition to the particulars from the story presented above, the following also holds: "Tybalt does not know that Juliet does not know that Tybalt read the letter," and specify a state of the world in your model that corresponds to the situation.
3. On day one, you are told an integer n . On day two, you are only able to remember that the number is either $n - 1$, n , or $n + 1$. Is it possible to use a knowledge space to model your information on day two? Why or why not?
4. Consider a knowledge space in which: $N = \{1, 2\}$, $\Omega = \{1, 2, 3, 4, 5, 6, 7\}$,

$$\mathcal{P}_1 = \{\{1\}, \{2, 3\}, \{4, 5\}, \{6, 7\}\}, \mathcal{P}_2 = \{\{1, 2\}, \{3, 4\}, \{5, 6\}, \{7\}\}.$$

- (a) Give an equivalent description of the knowledge space above in terms of type spaces T_i and signalling functions τ_i .
 - (b) Let $A = \{1\}$ and $B = \{1, 2, 3, 4, 5, 6\}$. Identify the events K_1A , K_2A , K_2K_1A , K_1K_2A , K_1B , K_2B , K_2K_1B , K_1K_2B , $K_1K_2K_1B$, $K_2K_1K_2B$.
5. Consider a knowledge space in which $N = \{I, II\}$, $\Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$,

$$\mathcal{P}_I = \{\{1\}, \{2, 3\}, \{4, 5\}, \{6\}, \{7\}, \{8, 9\}\}, \mathcal{P}_{II} = \{\{1\}, \{2, 5\}, \{3\}, \{4, 7\}, \{6, 9\}, \{8\}\}.$$

What are the connected components of the graph G_C ? Which events are common knowledge in state of the world $\omega = 1$? Which events are common knowledge in state of the world $\omega = 9$? Which events are common knowledge in state of the world $\omega = 5$?

6. Construct an example of a knowledge space with a set of players $N = \{1, 2, 3\}$ and an event A that is not common knowledge among all the players N , but is common knowledge among players $\{2, 3\}$.

Math corner

7. Prove that if events A and B are common knowledge among the players at state of the world ω , then the event $A \cap B$ is also common knowledge among the players in ω .
8. Given a knowledge space $B = (N, \Omega, \{\Sigma_i\}_{i \in N})$, find a knowledge space $B' = (N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$ such that each Σ_i is the algebra generated by player i 's information partition in B' .
9. Prove claim 1.13 in the lecture slides.
10. Prove claim 1.14 in the lecture slides.
11. Prove theorem 1.20 in the lecture slides.¹
12. Prove that event A is common knowledge in every state of the world in A if and only if $K_1K_2 \dots K_nA = A$, where $N = \{1, 2, \dots, n\}$ is the set of players.

¹You should try this by yourself. If you can't you can check the lecture notes from Omer Tamuz from Caltech that I uploaded.